DESIGN BASELINE 0.3 · RESEARCH CUT 24 OCTOBER 2026 · MICRO CENTER-ONLY SINGLE-ZONE REVISION 24 OCTOBER 2026

Determine

Dumb glasses, a capable phone, and a calibrated warning about what is directly ahead of you.

Architecture and prototype contract

Status: proposed design. Nothing here is built, clinically validated, or cleared as a medical device. It does not replace a white cane, a guide dog or orientation-and-mobility training.

WEARABLE
Two dumb boards: an ESP32-CAM that streams JPEG, and a XIAO ESP32-C3 carrying one forward time-of-flight zone and the two temple haptic drivers. No AI on the frame, and no audio hardware on it either: cue sounds are synthesised on the phone and played through a finished bone-conduction Bluetooth headset.

PHONE
Does all perception, decides which hazard to report, runs calibration and the user interface. Android first.

USERS
Blind and low-vision travellers, including people with unilateral or asymmetric hearing loss.

TARGET
HackTX 26, Oct 24–25 2026, UT Austin. Build on site within the event rules.

> **Pitch.** When you reverse a car, your view is limited, so the car gives you parking sensors. People who are actually blind or have low vision mostly do without a head-level equivalent. Determine is that sensor for a walker: it beeps faster as an obstacle gets closer, places the warning on the left, centre or right, and calibrates its loudness so both sides are heard equally, even with hearing loss in one ear.

How to read this contract

This document fixes *interfaces and invariants*. It does not claim that the hardware works or that the design has been field-tested. **MUST**, **MUST NOT**, **SHOULD** and **MAY** state obligations on the proposed implementation only. Every claim carries a tag:

Tag	Meaning
FACT	Read from a primary or vendor source, cited with [n]. Vendor statements describe the vendor's own product, not an independent check of Determine.
DESIGN	An engineering decision this document makes, with its rationale stated.
INFERENCE	Arithmetic or reasoning from facts, not measured. A gate must confirm it before anyone repeats it as a result.
PROPOSED	A starting threshold to tune in a pilot. It is not a standard, a regulation or a measurement.
GATE	Measured evidence required before a capability may be claimed or the next layer built.

Procurement status, 24 Oct 2026 ~1:40 PM CDT: Houston stock and prices were checked in Zen (Chapter 12), not reserved. The motor supply is one short, the LM3671 is sold out, and integrated bone-conduction audio is unresolved. The separate OpenRun headset is not approved for this build; headset diagrams and calibration describe the implemented software/bench path, not completed integrated hardware. All new purchases remain Micro Center-only. Prices exclude tax and may change; phone/tools are owned only if confirmed. Determine is an independent project.

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Decisions at a glance

Question	Decision	Why
Where does computing happen?	On the phone. Two dumb boards on the glasses capture, transmit and buzz; the phone synthesises and plays the sound. DESIGN	Cheap, replaceable frame. The phone already has an NPU, a battery and accessibility settings. One place to update models.
Two cameras?	No, for v0. One RGB camera whose frames feed only the optional Describe path and frame diagnostics (no on-phone object detection), plus ONE single-zone short-range ToF (VL53L4CD on an Arduino Modulino Distance) for forward proximity. The camera never supplies range, and neither sensor supplies left/right. DESIGN	Stereo depth needs synchronised capture and calibration, and the second ESP32-CAM is a spare, not a stereo partner. The 8×8 grid of the earlier revision is not locally purchasable; a single zone measures forward distance class-agnostically inside a Micro Center-only budget.
Camera on the cue board?	No. A separate ESP32-CAM streams JPEG over the same hotspot. DESIGN	The XIAO ESP32-C3 has no camera, and the esp32-camera driver supports ESP32, S2 and S3 only [13][59]. A second Wi-Fi producer is the price of keeping the cue board simple.
Camera link	JPEG from the ESP32-CAM over the phone's Wi-Fi hotspot. USB UVC only after a per-phone check. DESIGN	BLE cannot carry useful video [15]. Android external-camera support depends on the phone maker [10].
Urgent cue path	The phone synthesises the cue sound and plays it over a finished bone-conduction Bluetooth headset; the same CUE command fires temple haptics on the C3 over UDP. No board audio. DESIGN	Dropping on-board I²S audio removes amplifiers, raw transducers and their shipped parts from the bill of materials, leaving one store for everything new. A2DP's 100–200 ms [16] lands only on the audio leg; urgency rides the millisecond haptic leg and the tempo grammar.
Conveying direction	Not claimed in v0. One forward zone has no lateral resolution, and a camera detection is never associated with the ToF return. The two calibrated audio channels and two temple motors remain for hearing calibration and future lateral sensing. DESIGN	Inferring left/right range from one zone, or attaching an unrelated camera object to it, would be fabrication. Redundant per-side channels survive in the calibration story (Chapter 9), not in hazard claims.
Calibration	Per session, per side: detection threshold, then a loudness balance, with a hard ceiling enforced by the phone's audio path. DESIGN	This is the user's core requirement. Coupling and hearing differ by person and by how the frame sits.

Question	Decision	Why
Cloud	Never in the hazard loop. Scene description is optional and labelled "may be wrong". DESIGN	Network loss must not silence warnings. A 2026 study found multimodal-LLM answers wrong 22.2% of the time [30].
Meta and AWS	Meta DAT is an alternative frame source behind the same interface. AWS is user-requested description only: a Graviton2 Lambda API plus SmolVLM on Inferentia2 is the design target, untested (§7.3). DESIGN	Shows real integration without weakening the cost thesis or the offline invariant.

CHAPTER 1

Mission, guarantees and non-goals

1.1 The precise product

Determine is a **secondary electronic travel aid** (ETA). It adds a near-field warning layer, focused on chest- and head-level obstacles, on top of the traveller's primary aid. The 2026 *Scientific Reports* study defines primary aids as the white cane and guide dog, and ETAs as supplementary tools [2]. A long cane covers the ground and "does not protect users from obstacles at the upper body and head levels" [2]. In a survey of 307 blind travellers, 13% had a head-level accident at least monthly, 86% of head-level accidents happened outdoors (mostly tree branches), and 23% of head-level accidents had medical consequences [3]. FACT

The wearable is deliberately "dumb": it has sensors, radios and actuators, but it makes no decisions. The phone app is the processing unit. It decides what counts as a hazard, which single cue to play and how loud each side should be. DESIGN

1.2 Guarantees

1. **G1 Fail loud.** If the camera, ToF, link, headset route, phone app or wearable power fails, the system targets an audible or tactile "system unavailable" notification through a surviving output within ≤ 1 s PROPOSED. The phone owns audio, so losing the headset route is itself a fault: it is announced explicitly (and the temples keep buzzing if the cue board is alive); the app never silently reroutes cue audio to the phone speaker. Silence means "nothing to report *from a working system*", never "path clear".
2. **G2 Unknown is not clear.** An empty detection list never produces a reassurance cue. Determine has no "all clear" sound.
3. **G3 Offline hazard loop.** Capture → perception → decision → cue runs with no internet connection.
4. **G4 Calibrated, capped output.** Per-side listening level comes from the wearer's own calibration and is clamped by a phone-enforced ceiling that no setting can exceed; the board separately clamps haptic intensity to `haptic_max`.
5. **G5 Redundant urgency.** Every warning travels on at least two channels the wearer has confirmed during calibration that they can perceive (headset audio plus temple haptics). Direction is not carried at all in v0, rather than carried badly.
6. **G6 Bounded vocabulary.** Hazard cues come from a small, learnable set (Chapter 8). Free-form speech never pre-empts a hazard cue.
7. **G7 Privacy by default.** Frames stay in phone memory. Images are not retained or uploaded unless the user explicitly turns on a separate feature.

1.3 Explicit non-goals

Not promised	Reason
Replacing the cane, guide dog or O&M training	Primary aids provide ground contact and legal recognition. ETAs supplement them [1][2].
Drop-offs, curbs, stairs down, holes	Forward-facing head-level sensing does not see the ground ahead reliably. No claim without per-class test data.
Street crossing or traffic judgement	Safety-critical. Out of scope.
Glass doors, thin wires, wet or very dark surfaces	Known failure modes for camera and ToF sensing, listed as "cannot see" in the app.

Not promised	Reason
Restoring directional hearing	Bone conduction and rerouting do not restore localisation in single-sided deafness [28].
A certified sound-exposure dose from bone conduction	No verified standard maps bone-conduction headset output to air-conduction dose limits. The prototype caps output but cannot certify it.
Unsupervised public trials during HackTX	Demos run on a staged indoor course with a sighted spotter.

1.4 Claim classes

Class	Meaning	Example
C0	Measured on the prototype, with the log kept	"p95 capture-to-cue latency was 212 ms over 120 trials." Only after the test exists.
C1	Vendor or primary-source fact about a part	"The VL53L4CD reaches 1200 mm indoors at 90% detection on a white target, 18° diagonal FOV" [20].
C2	Design target	"Unavailable cue within 1 s."
C3	Research direction	"Deliver cues to LE Audio hearing aids."

The pitch, slides and Devpost entry **MUST** label class C2 and C3 statements as such. Presenting a C2 target as a C0 result is a release blocker.

CHAPTER 2

Evidence base: travellers, hearing and cues

2.1 What a real ETA trial shows

Pittet et al. (2026) tested 13 blind and low-vision participants on an indoor corridor with floor, waist and head-level obstacles. They compared cane alone with cane plus NOA (a camera-based harness) and cane plus BuzzClip (ultrasonic) [2]. FACT

Measure (mean)	Cane only	Cane + NOA	Cane + BuzzClip
Body contacts with obstacles	2.92	1.62	3.54
Cane contacts	4.69	2.62	3.15

What this means for Determine:

- **Precision beats volume.** Participants rated NOA higher on precision and on upper-body detection. The BuzzClip's vibrations were criticised as too weak and too frequent, and gave no improvement in incidents [2].
- **Learnability matters.** 4 of 13 participants had trouble understanding NOA's spatialised warning sounds [2]. Determine therefore uses a small set of cues, trained during onboarding, with direction carried redundantly.
- **Short, precise beats generative.** For finding objects, every participant preferred concise output such as "Person to your left at 11 o'clock, 3 metres" from a local deep-learning model over generative-AI descriptions [2].
- **Test with the real population.** Of 140 ETA studies reviewed, only 29% included blind or low-vision participants [2]. A blindfolded sighted tester does not validate Determine.

2.2 The glasses premise, without stereotypes

Some blind people wear dark glasses, but many do not. Blindness is a spectrum, and many legally blind people use their remaining vision. Determine therefore offers clear, tinted and clip-on configurations, and its sensors mount on the frame front so that lens tint does not affect sensing. DESIGN A pair of glasses is a natural place for the hardware: it sits at eye level, it points where the head points, and it already rests on the temples, where bone conduction and haptics make contact.

2.3 Hearing: two different problems

Asymmetric hearing loss

Both ears hear, but one hears less. Per-side gain can equalise *perceived* loudness, which is exactly what the user asked for. Calibration is the right tool here. INFERENCE

Single-sided deafness (SSD)

One cochlea does no useful work. Every left or right channel ends up at the same working cochlea. In 28 unilaterally deaf subjects, sound crossing the skull lost only about 0–10 dB, depending on frequency, with roughly 40 dB variation between people [26]. Left/right audio therefore cannot carry direction, so haptics and spoken words must. FACT → DESIGN

- Bone-conduction headsets lateralise sound much like headphones in a small study. Even a 20 dB level difference gave only about 75% correct lateralisation [27]. Stereo bone conduction is useful, but it is not reliable enough to be the only carrier of direction. FACT

- In one study, a contralateral-routing bone-conduction device did not improve localisation for people with single-sided deafness [28]. Monaural listeners rely heavily on head-shadow level cues [29]. FACT
- Audio played through bone-conduction headphones increased localisation errors for real environmental sounds [52]. Determine keeps cues short and sparse so it does not mask traffic and voices. FACT → DESIGN

2.4 Auditory-display findings used in Chapter 8

- **Earcon guidelines** (Brewster et al.): vary timbre, register and rhythm, and use musical timbres rather than sine waves. Make rhythms differ substantially, including in note count. Keep pitch roughly within 125–150 Hz to 5 kHz. Notes should last at least 0.0825 s, at most about 6 notes per second, with about 0.1 s between serial earcons. Keep intensity 10–20 dB above threshold. Do not rely on register or intensity alone for absolute identification [24]. FACT
- **Spearcons**, which are sped-up spoken words, beat earcons and auditory icons on menu-navigation efficiency and accuracy, and are learned about as fast as speech [25]. FACT
- **Safe listening:** WHO lists 80 dB as safe for up to 40 hours a week and 90 dB for only 4 hours. It recommends keeping devices at or below 60% of maximum volume [4]. The WHO-ITU standard defines exposure modes for adults and sensitive users [5]. FACT

CHAPTER 3

Landscape and positioning

Product	Form	Relevant capability	Price (source, 24 Oct 2026)	Limitation relative to Determine's goal
Ray-Ban Meta (Gen 3)	Camera and open-ear speaker glasses	Meta AI scene description, Be My Eyes volunteer calls. Developer access via Wearables Device Access Toolkit (DAT) 1.0 [6]	From $449 [42]	AFB calls it an accessibility tool "only by coincidence" and reports hallucinated object positions [9]. Air-conduction speakers, not bone conduction. No haptics. Gen 3 is not named in the DAT supported-device list [6]
Envision Ally Solos	Camera glasses, phone-tethered	Reading, scene description, faces	$699 shop listing [43]	Description-focused, not near-field hazard cues
OrCam MyEye 3 Pro	22.5 g clip-on camera	Reading, faces, products	$4,250 reseller [44]	Reading-centric. Not a navigation device
WeWALK Smart Cane 2	Smart cane handle	Chest and head-level obstacle detection with vibration, navigation app	Not on the static official page [45]	Occupies the cane hand. Closest neighbour in function
biped NOA	Shoulder harness, three depth cameras	Spatialised obstacle audio sent to bone-conduction headphones [2]	$4,990 reseller [46]	8 of 13 trial users found it heavy or bulky [2]. Expensive
.lumen glasses	Head-worn, forehead haptics	Haptic path guidance	€9,999 [47]	Ships only in Romania per its order page. Claims such as "better than the guide dog" are unvalidated marketing
NaviLens	Markers plus a free app	Long-range marker reading in transit systems	Free [48]	Needs installed infrastructure. Wayfinding, not obstacle avoidance
Seeing AI / Lookout	Phone apps	Text, scene and object description	Free [49][50]	Handheld. Lookout's Explore mode is beta and "less accurate" [50]
Microsoft Soundscape	Retired spatial-audio app	3D audio beacons	Open-sourced [51]	Microsoft says it is "not … for use as a mobility aid" [51]

3.1 Where Determine sits

None of the products found combines all four of the following: a low-cost glasses form, phone-side perception, bone-conduction *and* haptic output, and explicit calibration for asymmetric hearing. No ETA product or paper found addresses unilateral hearing loss in its cue design. That is an inference from absence of evidence, not a verified gap. INFERENCE

Not "AI glasses for the blind"

That framing is crowded and invites overclaiming. Determine is a *parking sensor for walking*, whose output follows a defined set of rules.

Commodity hardware, one store

Two boards, one forward ToF zone and two intended temple haptics. Houston stock was checked in Zen; the motor shortage, sold-out regulator and unresolved integrated audio prevent a complete-build purchase claim (Chapter 12).

Hearing-aware by construction

Per-side calibration, a mono-safe mode for single-sided deafness and haptic confirmation are core features, not settings bolted on later.

> **Claims to avoid in the pitch:** "replaces the cane", "detects stairs, curbs or traffic", "makes navigation safe", "restores hearing", "first obstacle sensor for the blind" (WeWALK and NOA already exist), "uses Auracast" (unless it was actually built and demonstrated).

CHAPTER 4

System architecture and seams

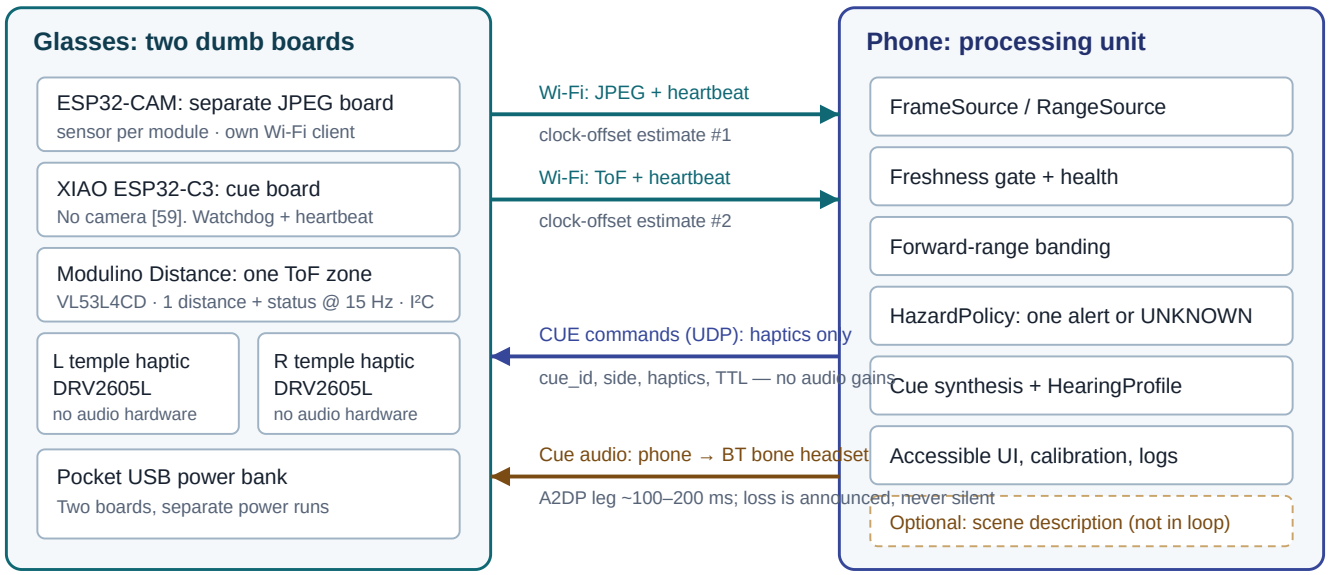


Figure 4.1. Topology: two independent Wi-Fi producers and no board-to-board bus. The phone is the only integrator — it aligns each stream by that board's estimated clock offset and receive age, and discards stale input. The wearable never decides what is hazardous. Cue audio is synthesised on the phone and played over a finished bone-conduction Bluetooth headset; the wearable only plays haptics. Cue commands carry a time-to-live (TTL), so a stale command is dropped rather than played late.

4.1 Module seams

Each seam exists only where at least two implementations are real or planned, following the "two adapters means a real seam" rule. DESIGN

Module	Interface	Adapters	Invariants and error modes
FrameSource	`next(): Frame{seq, tCapture, w, h, jpeg}` or `Unavailable(reason)`	ESP32-CAM Wi-Fi · USB UVC · Meta DAT · phone camera (Tier 0)	Sequence numbers increase monotonically. Clock offset is estimated per board by PING/PONG; receive age is tracked. Stale input is discarded. Disconnection is explicit, never an empty frame.
RangeSource	`next(): RangeSample{seq, t, mm:u16, status:u8}`	VL53L4CD (Modulino Distance) on the XIAO ESP32-C3 · none (camera only)	Canonical status: 0 VALID, 1 NO_TARGET, 2 UNKNOWN (raw VL53L4CD 0 is valid; invalid/error maps to UNKNOWN, never to NO_TARGET). An invalid sample means UNKNOWN, never "far". Same per-board offset and staleness rules as frames.

Module	Interface	Adapters	Invariants and error modes
Frame use	`latest(): Frame` for Describe and diagnostics	Describe client with on-phone face redaction · judge screen frame age	Frames never enter the hazard decision: there is no on-phone object detector, no label and no bearing. Nothing from the image is attached to the ToF return, so the camera never gates, bands, ranges or lateralises a hazard.
HazardPolicy	`select(range, motion): Alert | Unknown | Quiet`	Single rule-based implementation (internal seam for tests)	At most one alert at a time, from the single forward sample. Entry and exit use hysteresis. Never emits "clear", never infers left/right.
CueSink	`present(alert, profile): Ack | Fault`	Phone audio to the BT bone headset · temple haptics over UDP · phone vibration (non-directional)	Audio levels are clamped by the phone-side profile ceiling; haptic intensity is clamped on the board to `haptic_max` (ACK APPLIED_CLAMPED). A missing acknowledgement escalates to UNAVAILABLE. A lost headset route is announced explicitly, never silently rerouted to the phone speaker.
HearingProfile	`calibrate(sink): Profile` · `levels(cue): (gL, gR)`	One per output device	Calibration is timestamped and re-run each session. A side outside the ceiling is marked unusable. The ceiling lives in the phone's audio path; the board only clamps haptic intensity.

4.2 Runtime state machine

```
STARTING ──(frames fresh ∧ ToF ok ∧ cue ack ∧ (HAPTIC_ONLY ∨ headset route confirmed))──▶ READY
STARTING ──(links and outputs confirmed ∧ fresh UNKNOWN sample)──▶ DEGRADED   # no false startup timeout
READY ──(ToF UNKNOWN ∨ range stale)──▶ DEGRADED   # stop metric warning repeats, announce limitation
READY/DEGRADED ──(frame age > 500 ms ∨ no heartbeat > 500 ms from either board ∨ repeated cue failure ∨ (audio enabled ∧ headset route lost) ∨ phone battery critical)──▶ UNAVAILABLE
UNAVAILABLE: announce "system off" on surviving outputs — headset audio if the route is alive, temple haptics if the cue board is alive, phone vibration; suppress all obstacle cues; retry link. Losing the headset route is itself announced, never silently handed to the phone speaker
any ──(user pause)──▶ PAUSED   # explicit, spoken confirmation; never automatic
```

The 500 ms frame-age and per-board heartbeat timeouts are **PROPOSED**; heartbeat interval targets 100 ms. They leave time inside the untested ≤ 1 s fail-loud goal and must be tuned at Gate G7. A bank without telemetry cannot report low charge before its outputs die.

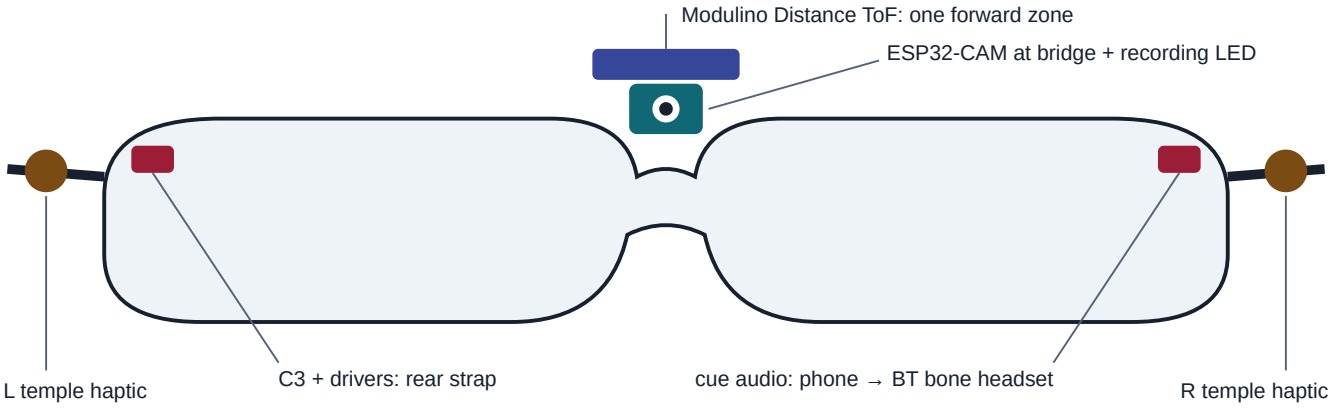
4.3 Wire contract (v2)

```
UDP over the phone hotspot · little-endian · header 8 B {magic 0x5444, ver=2, type, seq:u32} ·
receivers drop wrong magic/ver/type/length
ESP32-CAM → phone
  FRAME   {frame_id:u32, t_cap_us:u64, w:u16, h:u16, total_len:u32, offset:u32,
           chunk_len:u16, chunk_idx:u16, chunk_count:u16, data}          # ≤ 1438 B, JPEG chunks
  BEAT    {board, reset_reason, faults:u16, t_us:u64, rssi, temp, heap_kb}
XIAO ESP32-C3 (cue board) → phone
  RANGE   {t_us:u64, mm:u16, status:u8}                                  # 11 B body, 19 B total
  ACK     {cue_seq:u32, status:u8, reserved:u8, t_board_us:u64}          # 14 B body, 22 B total
  BEAT    {…}
Phone ↔ each board   (clock estimation; the camera board accepts no cue commands)
  PING    {t_phone_us}  →  PONG {t_phone_us, t_board_rx_us, t_board_tx_us}
Phone → cue board
  CUE     {t_issued_phone_us:u64, deadline_board_us:u64, cue_id:u8, side:u8,
           haptic_pattern:u8, haptic_l:u8, haptic_r:u8, flags:u8, ttl_ms:u16}   # 24 B body, 32 B
total; no gain fields
  PROFILE {mode:u8, haptic_max:u8}                                       # 2 B body, 10 B total;
haptic_max 1..255 (0 is BAD_CUE)
RANGE status: 0 VALID · 1 NO_TARGET · 2 UNKNOWN   (raw VL53L4CD 0 = valid; invalid/error → UNKNOWN,
never NO_TARGET)
ACK status:  0 APPLIED · 1 EXPIRED · 2 APPLIED_CLAMPED (haptic_max lowered a requested intensity; side
exclusion alone is not clamping)
             · 3 NO_SYNC · 4 BAD_CUE (side > 2, unknown haptic_pattern, PROFILE mode > 2, or PROFILE
haptic_max = 0) · 5 BUSY_DROPPED
PROFILE mode: 0 STEREO · 1 SSD · 2 HAPTIC_ONLY — phone audio-route semantics only; the board is haptic-
only regardless of mode
PROFILE haptic_max: 1..255. Zero is rejected so the watchdog LSS, now the last local failure channel,
can never be configured silent. 1 is a protocol floor, not a measured guarantee the wearer can feel it
faults: bit2 reserved (v1 I²S fault removed; the board has no audio hardware)
v1 → v2: audio gains left CUE/ACK entirely — levels are phone-side; RANGE shrank from a 64-zone grid to
one sample
```

There is no board-to-board bus: the phone is the only integrator. The phone estimates a separate clock offset for each board, timestamps receipt, and discards `FRAME` or `RANGE` whose offset-corrected capture time or receive age exceeds the freshness bound. The cue board compares `deadline_board_us` against its offset-corrected current time, dropping a late `CUE` (ACK EXPIRED) rather than resetting its lifetime on arrival; `deadline_board_us = 0` means no sync (ACK NO_SYNC). One CUE drives both legs of one cue: the phone synthesises the sound for the headset while the board plays the haptic pattern and acknowledges. The C3's watchdog runs haptics only — LSS on both temples, capped at `haptic_max` — when phone PINGs stop while the board is powered; the phone announces a dead board explicitly on its own outputs. Clock uncertainty, command expiry and both failure directions must be tested in Gates G6–G7. Power-bank battery percentage is not reported by either board unless external sensing is added; loss of its power is detected by missing heartbeats on the phone. DESIGN

CHAPTER 5

Wearable hardware



Only the camera and the single-zone ToF face forward; the C3 and drivers sit on a rear strap.
Cue audio plays from the phone; the power bank stays in a pocket.

Figure 5.1. Proposed compact headband layout, not finished eyeglasses. The Modulino Distance board carries one VL53L4CD zone — 1200 mm at 90% detection indoors, 18° diagonal FOV [20][39]. It is a near-field demo sensor, not a navigation rangefinder. The only audio hardware is the finished bone-conduction headset, driven from the phone. Weight is kept off the nose. Fit, cable routing and comfort remain untested.

5.1 Sensing options

Configuration	What it gives	Cost and risk	v0?
Mono RGB only	Object classes and bearing (left/centre/right) from image position	No distance for arbitrary obstacles. Common detectors cover objects such as people, chairs and cars, but not branches or poles INFERENCE	Fallback
Mono RGB + single-zone ToF	Class-agnostic forward proximity, one distance at a time: "how far ahead", never "where left/right". VL53L4CD (Arduino Modulino Distance): 1200 mm at 90% detection indoors on white (≈550 mm in overcast daylight, ≈450 mm on gray), 18° diagonal FOV, up to 100 Hz, Class 1 940 nm laser [20][39]. Camera frames serve Describe and diagnostics only	SKU 992826 [40]: Houston website stock 2, $11.99 at the §12 snapshot. Glass, dark surfaces and strong sun remain unmeasured	**Yes**
Dual RGB stereo	Dense depth after calibration	Needs synchronised exposure and stereo rectification. The ESP32-CAM carries one sensor and the C3 has no camera interface at all [13][59]; the second camera in the pair is a spare, not a stereo partner	No
Mono + learned depth	Relative depth. Depth Anything V2-Small is Apache-2.0, while its larger variants are non-commercial [23]	No metric scale unless anchored to the ToF. Phone compute load	Stretch
Stereo depth module (e.g. OAK-D Lite)	On-device depth	Expensive, multi-watt, harder on Android (secondary source only)	No

Configuration	What it gives	Cost and risk	v0?
Meta Ray-Ban via DAT	Glasses camera stream up to 720×1280 at 2–30 fps over Bluetooth [6][58]	No ToF or haptics. Latency undocumented. Requires owning the glasses. Gen 3 not listed [6]	Optional adapter

Why one forward ToF zone instead of a grid or a second camera? Head-level hazards such as branches, sign edges and open cabinet doors are rarely classes a detector knows, so the hazard decision must not depend on recognising the object, and ToF measures distance regardless of what the object is. The earlier revision's 8×8 grid is not locally purchasable under the Micro Center-only rule; the single zone keeps class-agnostic distance but gives up corridors, flanks and any left/right resolution — the hazard policy (§7.2) is a forward proximity band, and nothing from the camera is attached to the ToF return. DESIGN

5.2 Wearable board choice (two-board revision, 24 Oct 2026; Micro Center-only 24 Oct 2026)

The XIAO ESP32-S3 Sense is not obtainable for this build, so the wearable splits into two boards that share no bus: a **cue board** (single-zone ToF over I²C and two haptic outputs; no audio hardware) and a separate **camera board** that only streams JPEG. Both join the phone's hotspot as independent clients (Chapter 4). Both boards are on the §12.2 purchase list (required unless the team already has them). DESIGN

Board	Relevant facts	Verdict
Seeed XIAO ESP32-C3 (3-pack)	RISC-V single core, 400 KB SRAM, 4 MB flash, Wi-Fi b/g/n + BLE 5, one I²C and one SPI bus, 11 GPIO, 21×17.8 mm, no camera [59]. Vendor-typical Wi-Fi current 75 mA [59]	**Selected cue board.** SKU 069005: Houston 3 packs, $15.59 per pack at §12 snapshot. One used, two spares
ESP32-CAM (classic ESP32)	Espressif supports ESP32, S2 and S3; PSRAM is required except JPEG at CIF or lower [13]	**Selected camera board.** SKU 221739: Houston 6 pairs, $17.99 per pair at §12 snapshot. One used, one spare, not stereo. Verify actual PSRAM and sensor at G0
Seeed XIAO ESP32-S3 Sense	8 MB PSRAM, camera board (OV3660 or OV2640 since 2025-06-30), microphone, one DVP camera interface [12] [14]	**Not selected:** unavailable to this build. Documented single-board alternative for a later revision ($13.99 online when checked 24 Oct [14])
Raspberry Pi Zero 2 W	Quad A53, CSI-2, H.264 1080p30 encode, Wi-Fi and BT 4.2, 5 V/2.5 A supply spec [53]	Overkill for a "dumb" frame. Power and boot time. Out of stock at Adafruit
Raspberry Pi Pico 2 W	RP2350, 520 KB SRAM, no camera port [54]	Cannot carry the camera role
ESP32-P4	USB 2.0 high-speed, MIPI-CSI with ISP, H.264 up to 1080p30, but no radio [55]	Roadmap: wired high-speed or Wi-Fi via a companion chip

5.3 Output hardware

- **Cue audio: implemented headset path, integrated hardware unresolved.** The phone synthesises cues and supports Bluetooth headset playback with per-side ceilings and explicit route-failure handling; it never silently falls back to its speaker. The separate Shokz OpenRun is not approved for the intended build and its old listing redirects to support [41]. No suitable stocked discrete bone-conduction transducer was identified; the Humbird listing says “No longer carried”. No raw-transducer/amplifier replacement is selected. Classic Bluetooth latency [16] remains a bench-path consideration, not proof of final hardware. OPEN DESIGN GATE
- **Haptics.** Two temple motors, one per side, each driven by a DRV2605L (§12.2: DRV2605L ×2, ERM motor ×2, TCA9548A mux ×1). Both drivers share I²C address 0x5A [36], so the TCA9548A mux isolates them on separate

channels; the VL53L4CD keeps its own default address on the upstream bus [20]. In v0 both temples fire together for every forward hazard — the single zone has no lateral resolution — while per-side intensity remains available for calibration verification and future lateral sensing. The board clamps requested intensity to the profile's `haptic_max` (1–255) and reports APPLIED_CLAMPED when it lowered a value (§4.3). DESIGN

5.4 Power

The C3 and Modulino use one USB-C feed; the ESP32-CAM gets a separate 5 V run from a pocket power bank. Common signal grounds are required. The listed Energizer bank supports simultaneous charging, but measured combined draw, brown-outs, thermal load and fit remain G0 checks. The LM3671 is sold out; TPS62827 is an unadopted 3.3 V rail candidate (3.4–5.5 V input), not the same pinout. An extra rail may be omitted only if G0 measurements support it. Integrated audio power requirements are unresolved.

GPIO budget: the C3 has one hardware I²C bus [59] for the Modulino Distance and the TCA9548A; the two DRV2605Ls sit on separate mux channels. No I²S pins are needed in v2. Fix the complete pin map and the status-LED GPIO before soldering; do not power the motors from an unmeasured C3 3V3 output. GATE G0

Rail assignment is a G0 decision: Modulino/mux logic from a verified C3 3V3 budget or a verified separate regulator; DRV2605L supply within vendor limits. Do not assume a regulator substitution or motor-drive redesign. The untested `DT_HAPTIC_MOSFET` flag has no selected shopping-list parts.

CHAPTER 6

Links: Wi-Fi, USB, BLE, LE Audio, Auracast

6.1 Camera and sensor transport

Link	Facts	Use in Determine
Wi-Fi to the phone hotspot	Both wearable boards join the hotspot as stations: the ESP32-CAM streams JPEG and the C3 sends ToF samples and acknowledgements. esp32-camera recommends JPEG, because YUV and RGB strain PSRAM bandwidth when Wi-Fi is on [13]. Both boards and the phone share one hotspot channel's airtime	**Primary.** QVGA–VGA JPEG at 10–15 fps target. Joining the phone's hotspot keeps the phone's cellular internet, which joining a glasses-hosted access point would not. INFERENCE Effective throughput with two stations is unmeasured; Gate G1 measures it
USB-C UVC	Android supports UVC webcams through Camera2, but only if the device maker enables the external camera provider. AOSP says it "isn't designed" for high-speed streaming or AR, and its sample configuration bounds 640×480 at 30 fps [10]	Secondary, only after a per-phone check with a UVC camera. A user-space UVC library over the Android USB host API [11] is a further fallback
BLE GATT	ESP32-to-ESP32 throughput is about 0.73 Mbit/s on the 1M PHY and about 1.4 Mbit/s on the 2M PHY. Phone throughput is lower and undocumented [15]	Provisioning, health, and possibly cues. Not video
Meta DAT	The camera stream degrades resolution first, then frame rate, and does not drop below 15 fps. A2DP and HFP audio are mutually exclusive [6][7]	Alternative FrameSource for teams that own supported glasses

6.2 Audio paths and latency

Path	Latency facts	Role
Phone → Classic BT bone headset (A2DP)	Typically 100–200 ms [16]	**Primary cue audio.** Spoken detail and cue identity; urgency rides the temple haptics, not this leg
Phone → LE Audio unicast (LC3)	LC3 uses 10 ms frames. Best case is "just over 20 ms" end to end. Every receiver must support a 40 ms presentation delay, and hearing-aid live use stays at ≤ 20 ms [16]. Android has had LE Audio since Android 13 [19]	Roadmap: deliver cues to LE Audio hearing aids and earbuds
Auracast broadcast	One-to-many, connectionless. The transmitter never knows who is listening [17]. Android support is limited to some devices (Pixel 8 and later, Galaxy S23 and later, some Xiaomi) [18]	Demo extra: judges listen on their own Auracast earbuds through an encrypted stream. Never the personal safety path
Phone → CUE (UDP) → temple haptics	LAN hop plus DRV2605L start. Unmeasured	**Primary urgent path.** Haptic only in v2 — the board has no audio hardware

> **Why the phone plays the earcons.** The "dumb glasses" principle is about where decisions are made, not about where speakers live. Moving audio to the phone deletes the amplifiers, raw transducers, per-side gain plumbing and shipped parts from the bill of materials, keeps every new purchase in one store, and puts level control where Android already exposes accessibility volume and mono settings. The cost is A2DP latency on the audio leg [16]; the design pays it deliberately and carries urgency on the millisecond haptic leg instead. DESIGN

6.3 End-to-end latency budget

Stage	Budget	Basis
Exposure + on-chip JPEG encode	≤ 60 ms	PROPOSED
Wi-Fi transfer (~30 KB frame)	≤ 40 ms	~1 MB/s effective throughput assumed for one station; the C3 shares the hotspot's airtime, so this is unmeasured for two producers. INFERENCE
Band + policy on phone (ToF only)	≤ 80 ms	Single-sample banding and policy; the camera is not in the hazard loop. Benchmark on the actual phone
CUE dispatch + haptic start on the C3	≤ 40 ms	UDP on LAN plus DRV2605L start. INFERENCE
Range-sample-to-haptic p95 target	**≤ 250 ms**	**Unmeasured with two Wi-Fi producers. Audio reaches the headset a further 100–200 ms later (A2DP) [16]; report the two legs separately**

At a walking speed of 1.4 m/s, every 100 ms of latency is about 14 cm of travel. INFERENCE So 250 ms is about 35 cm before human reaction time is even counted — and the sensor itself stops reporting beyond roughly 1.2 m indoors [20]. That is why this is a staged near-field demo with a spotter, not a validated warning distance: the FAR band's upper edge equals the sensor's spec limit, and latency plus reaction time consume a large share of even that margin.

CHAPTER 7

Perception and hazard policy

7.1 Pipeline

1. **Ingest and freshness gate.** Two independent producers: the ESP32-CAM's frames and the cue board's ToF samples each carry their board's clock, a per-board offset estimated by PING/PONG, and the phone's receive timestamp. Drop any input whose offset-corrected capture time or receive age exceeds the freshness bound. Every stage records a timestamp, for Gate G6.
2. **Forward range (class-agnostic).** Only raw VL53L4CD status 0 yields VALID and a distance; every other raw status maps to UNKNOWN. This includes empty or out-of-range views: the sensor cannot distinguish them from a weak-signal target. The board and simulator never emit the protocol's NO_TARGET value. UNKNOWN means no supported distance, never far or clear; it enters announced DEGRADED and stops metric warning repeats. There are no zones, corridors, flanks or left/right range.
3. **Camera frames (not in the hazard loop).** Frames are reassembled and kept only for the optional Describe path and the judge screen's frame age. There is no on-phone object detector, so no label, class or bearing is produced, and one forward zone could not be tied to a particular object in the frame anyway.
4. **Proximity band.** FAR / MID / NEAR come from the single forward distance when it is valid; beyond 1200 mm nothing is reported, and that silence is not a clear path. Never speak metres without a valid sample.
5. **Policy.** Pick at most one alert, with persistence and hysteresis (below).

7.2 Hazard policy (v0 rules)

```
inputs: range {mm, status}, head_yaw_rate
band(d): NEAR if d < 500; MID if d ≤ 899; FAR if d ≤ 1200; else NONE      # PROPOSED mm, ≤ 1200 mm
sensor spec; measured/tunable
usable:  VALID gives a distance; UNKNOWN stops the current alert and feeds DEGRADED, never a distance
or "clear"; NO_TARGET is a wire semantic not emitted by this board
enter alert:  same band ≥ FAR for ≥ 3 consecutive VALID samples (~200 ms @15 Hz)
exit alert:   band NONE (VALID beyond 1200 mm, or NO_TARGET) for ≥ 5 samples (re-enter with new cue)
rate limit:   ≤ 1 alert onset / 700 ms; repeat cadence set by band (Chapter 8)
suppress:     head_yaw_rate high (scanning) → hold FAR cues; NEAR and MID always pass
label:        always the generic "obstacle" — no detection class, no side
never:        output CLEAR; speak distance without a VALID sample; attach a camera detection to the
return; infer left/right; let description pre-empt NEAR
```

The bands are PROPOSED: FAR 900–1200 mm, MID 500–899 mm, NEAR under 500 mm, all measured and tunable at Gate G8. At 1.4 m/s, 250 ms of system latency plus about 1 s of human reaction consumes about 1.75 m of walking — more than the sensor's 1.2 m reach — so even FAR is a last-moment cue and this build supports a slow, staged near-field demo only. The 1.2 m figure is the vendor specification for a white indoor target [20]; it is not a validated useful warning distance, and gray or dark targets and bright sun shorten it (§5.1). INFERENCE One forward zone has no elevation resolution, so there is no low-return filtering, head-versus-chest zoning or mounting mask: it reports one distance for whatever is in its cone, and the cane's ground coverage is not modelled.

7.3 Optional scene description (Describe loop)

The user asks with a long press or voice: "What's ahead?" — never automatically. If the user has opted into cloud description and the service is available, the phone redacts faces and sensitive detail before upload; imperfect redaction is a privacy risk, so upload happens only on the user's explicit choice. A small arm64 Lambda API on Graviton2 [65] holds the prompt — two sentences, surroundings first, no guesses about people — and a small vision-language model describes the image: SmolVLM via Hugging Face Optimum Neuron on one Inf2 instance [62]

[63]. Optimum Neuron lists SmolVLM, and Gemma3 as a second candidate, for image-text-to-text on Inf2 with vLLM support [62], but neither deployment is **tested by us**. Trainium2 is the documented alternative [60][61][64], and Bedrock Nova Lite is a fallback that does not guarantee the silicon [66]; new AWS accounts start with zero Inf/Trn quota, so accelerator access must be requested early [67]. If no description service is ready, the phone says description is unavailable; it does not improvise an untested local model. A returned answer is prefixed "may be wrong". MID or NEAR hazards interrupt it, and it is inactive while the system is UNAVAILABLE. Response time is unmeasured; the design plans for seconds, not milliseconds. DESIGN

Implementation: contract `cloud/API.md`; API `cloud/describe-api` (Lambda arm64, SAM); server `cloud/inference` (optimum-neuron from git main on the Neuron SDK 2.31 PyTorch 2.9 AMI, TP 2 on inf2.xlarge, 512 px single tile). Only the CPU development path has been run: SmolVLM-256M answered in p50 2.0 s on a laptop CPU. Inferentia2 compile time and latency are unmeasured until quota is granted. GATE

7.4 "Cannot see" list (spoken during onboarding)

Glass doors and windows · thin wires and ropes · drop-offs, curbs and stairs going down · anything outside one narrow forward cone (about 18° diagonal [20]) · which side an obstacle is on · what the obstacle is · fast-moving vehicles · glare, darkness, rain, fog and bright sun · gray or dark surfaces at the edge of range · anything beyond about 1.2 m. Silence is not a clear path.

CHAPTER 8

The cue language

The requirement is a sound that is granular, easy to identify and level-aware. Determine treats the cue as a small grammar with four dimensions, so that no single dimension, and especially not loudness alone, has to carry everything. In v0 the "where" dimension is carried only by calibration and test cues: the single forward ToF zone cannot say left or right, so every hazard cue is a centred, both-sides cue. Brewster's guidelines warn against identification by register or intensity alone [24]. FACT → DESIGN

Dimension	Encoded by	Values	Rationale
What (family)	Timbre + rhythm motif	Obstacle: short wood-block double; at NEAR, one sharper urgent note. System: descending three-note. Ready: single soft tick, once. The person cue IDs (two-tone marimba) stay in the bank for phone sound semantics but nothing in v0 raises them from a ToF return	Brewster: timbre and rhythm are the strongest discriminators [24]
Where (side)	Centre only for hazards: both headset sides + both temples	Hazards: Centre. Left and Right survive only for calibration tones and the balance step	No lateral range exists to report. Redundancy is headset audio plus haptics, not left versus right [26][28]
How close	Repetition rate (parking-sensor metaphor)	FAR 900–1200 mm: 1 per s · MID 500–899 mm: 2.5 per s · NEAR under 500 mm: 5 per s, one urgent note each	Matches the car-sensor mental model. Peaks at 5 notes per second (FAR 2, MID 5, NEAR 5), within about 6 per second [24]
How urgent	Limited level step + haptic intensity	Base at calibrated sensation level. NEAR at +6 dB *within the phone-side ceiling*	Level supports urgency but is never the only signal. Stays 10–20 dB above threshold [24]

8.1 Timing grammar

```
note ≥ 83 ms (motif notes)   gap between serial cues ≥ 100 ms   max 6 notes/s      [24]
pitch range 300 Hz – 4 kHz  (inside Brewster 125 Hz–5 kHz; recheck against the headset's real response)
FAR:  ■■ ............ ■■ ............  (1/s, 2 notes/s)
MID:  ■■ ... ■■ ... ■■ ...            (2.5/s, 5 notes/s)
NEAR: ■ ■ ■ ■ ■ + haptic buzz         (5/s, one urgent note per 200 ms, +6 dB within the phone-side
ceiling)
SYSTEM OFF: ■ □ □ descending, repeated every 5 s until resolved, on headset audio, temple LSS haptics
and phone vibration
```

Each period must hold its notes plus the 100 ms gap: MID 2 × 83 + 100 = 266 ms of a 400 ms period; NEAR 83 + 100 = 183 ms of a 200 ms period. A multi-note motif cannot fit at 5 per second.

The implemented phone cue bank is a software/bench path; the final integrated audio hardware and its frequency response remain unresolved. Recheck pitch and level against whichever output is actually approved and measured.

8.2 Spoken layer

Spearcons are sped-up words such as "obstacle" or "near". They are optional and follow the earcon, never replace it. They are on by default during training and become opt-in once the wearer is fluent. Words never name a side, an

object class or a distance the sensor cannot back: with one forward zone a spoken distance (for example "1 metre") is allowed only from a VALID sample, rounded coarsely, and never paired with a camera label. DESIGN

8.3 Haptic mapping

Event	Left motor	Right motor
Obstacle FAR / MID	Both pulse together at band rate (pulse 40 ms / 60 ms)	
Obstacle NEAR	Strong 150 ms buzz on both temples, then band pulses	
System unavailable / board watchdog	Long-short-short (LSS) on both sides; the watchdog does this alone, with no audio, plus phone vibration when the phone is alive	

The two motors exist for the calibration balance step and future lateral sensing; in v0 the single forward zone fires them together. Haptics are the *urgent* channel and the confirming channel for headset audio. The BuzzClip result warns that weak, frequent vibration can be confused with cane-tip feedback [2], so intensity and cadence are user-tunable (capped by `haptic_max`) and validated in Gate G9. DESIGN

8.4 Cue bank as data

```
cue_bank  (synthesised on the phone; the same cue_id also rides each CUE packet to the cue board for
haptics)
{ "obstacle_far":  {"cue_id":1, "period_ms":1000, "haptic":"pulse40"},
  "obstacle_mid":  {"cue_id":2, "period_ms":400,  "haptic":"pulse60"},
  "obstacle_near": {"cue_id":3, "period_ms":200,  "haptic":"buzz150","level_step_db":6},
  "person_*":      {"cue_ids":[4,5,6], "note":"kept for sound semantics; not raised from ToF in v0"},
  "system_off":    {"cue_id":7, "period_ms":5000,"haptic":"lss","both":true} }
```

CHAPTER 9

Calibration for asymmetric hearing

Goal. Both sides of the cue should be *equally perceptible* to this wearer, with this frame position and this output device, without ever exceeding a hard ceiling. This is perceptual matching. It is not a hearing test or a diagnosis, and it does not restore hearing. DESIGN



Figure 9.1. Calibration flow, roughly 2–3 minutes and fully operable with TalkBack and screen-reader gestures.

1. **Fit and device.** The app asks the wearer to seat the headset and requires the Bluetooth bone-conduction route to be the active cue output; if it is not, calibration refuses to start rather than playing tones from the phone speaker. It records the output device and the wearer's self-reported weaker side. It warns about ambient noise if the phone microphone reads high.
2. **Per-side detection threshold.** An adaptive 2-down-1-up staircase, which converges near 70.7% correct [57], starts from a quiet level. The wearer answers "heard / not heard" by tapping. It runs in three bands that match the cue timbres (for example 500 Hz, 1 kHz and 2 kHz) and stops after 6–8 reversals. PROPOSED
3. **Presentation level.** Per band and side, the level is threshold plus a sensation level chosen by the wearer between 10 and 20 dB [24], clamped to the ceiling.
4. **Loudness balance.** Alternate the real "obstacle" cue left and right. The wearer nudges a balance control until both sides feel equally loud, which corrects for cue spectrum and for the bone crossing the skull. Repeat once to check consistency.
5. **Verify and choose a mode.** Play five random-side calibration cues and have the wearer name the side. Mode values are phone-side audio semantics (wire `mode`: 0 STEREO, 1 SSD, 2 HAPTIC_ONLY); the cue board is haptic-only in every mode.
   - **Stereo mode:** at least 4 of 5 correct and no side at the ceiling.
   - **SSD mode:** one side cannot be detected at the ceiling, or side naming is at chance. Audio becomes identical on both sides (mono). Hazard cues carry no direction in v0 anyway, so SSD costs nothing in the demo; any future lateral information would have to ride haptics and spearcons.
   - **Haptic-only mode:** for wearers who prefer it or cannot use audio. No headset is required and warning sounds are off; temple haptics carry warnings. Phone vibration and the explicit system-off alarm remain failure signals. Hearing calibration still requires a headset.

If the headset disconnects or rejects a tone, calibration enters a failed state, discards its result and requires reconnection and a fresh session. An unplayed tone is never scored as unheard, and a failed or disconnected session cannot apply a profile.

9.1 Profile and safety rules

```
HearingProfile {
  device_id, created_at, fit_note,
  mode: STEREO | SSD | HAPTIC_ONLY,
  ceiling_db,                     # phone-side; enforced in the audio path, never on the board
  bands: [{hz, thr_db_L, thr_db_R, sl_db, offset_db_L, offset_db_R}],
  haptic: {haptic_max, pattern_set},          # haptic_max is also sent to the board in PROFILE
  verified: {trials:5, correct:n}
}
```

- **Ceiling first.** No calibration step, slider or urgency step can exceed `ceiling_db` in the phone's audio path; the board separately caps haptic intensity at `haptic_max`, which the phone constrains to 1–255 — the haptic channel cannot be disabled because it is the last local channel left when the phone dies. If the weak side would need more, it is marked unusable. The app does not push more energy into it. The headset's own hardware volume remains outside the app's control, so the ceiling is relative to the volume set at calibration time. DESIGN
- **Re-run every session**, because how the frame couples to the head changes each time it is worn. A 20-second quick check suffices if the previous profile is under 7 days old. PROPOSED
- **Dose.** Track cumulative cue-on time. No final bone-conduction output has been selected or measured on an artificial mastoid; Determine cannot claim equivalence to WHO/ITU air-conduction limits [4][5].
- **Professional care.** The app recommends an audiologist for anyone with known hearing loss and never displays results as audiometry.
- **OS features stay reachable.** Android mono audio and balance settings remain available. Determine's profile works alongside them, not as a replacement.

CHAPTER 10

Safety, privacy and regulatory boundary

10.1 Threat and hazard ledger

Hazard	Mitigation	Residual risk
Missed obstacle (glass, wire, off-axis or beyond ~1.2 m, outside one narrow cone)	"Cannot see" list. Secondary-aid positioning. Cane mode. Staged near-field demo only	Inherent. Disclosed. One forward zone cannot give side or classify the object
False reassurance	No "clear" cue exists. UNAVAILABLE state. TTL on cues	Users may over-trust silence. Address it in training
Over-alerting and fatigue	One alert at a time, rate limit, persistence and hysteresis	Tune against measured false-alert rate
Masking environmental sound	Short sparse cues, open ears, cue-on duty tracked [52]	Measure in Gate G10
Excess vibration or sound level	Phone-side ceiling on audio; board `haptic_max` clamp on haptics. Session timer	No bone-conduction dose standard available; headset hardware volume is outside the app
Cue audio silently lost (headset disconnect, multipoint steal, wrong route)	Route check before READY and continuously; loss is announced on haptics and phone vibration; audio never silently falls back to the phone speaker	A2DP leg adds 100–200 ms; haptics carry urgency during the gap
Electrical and thermal	Pocket USB bank, no head-mounted Li-Po in v0, strain relief, temperature in heartbeat	Hackathon-grade wiring. Supervised use only
Bystander privacy	Visible recording indicator, no retention, demo signage and consent	Camera on face may still worry bystanders
Spoofed cue commands	v0 hotspot is WPA2 with a unique password. Roadmap: authenticated session key	LAN attacker during demo. Low impact, but logged

10.2 Regulatory boundary

21 CFR 886.5900 defines an *electronic vision aid* as a device that translates visual images of objects into tactile or auditory signals for a person with impaired vision or blindness. It is Class I and exempt from 510(k) premarket notification [31]. FACT Marketing Determine as a mobility aid would likely place it in this category. Being exempt from 510(k) does not mean being exempt from every FDA obligation, so get legal review of the remaining duties before any sale. The hackathon prototype is a research demonstration. INFERENCE

10.3 Human-subjects boundary

Testing with blind, low-vision or hearing-impaired participants beyond informal feedback SHOULD go through the university's human-subjects process, include an O&M specialist, and have informed consent and a spotter. Every HackTX demo uses a staged indoor course with a sighted spotter. DESIGN

10.4 Meta platform rules (if the DAT adapter is used)

Meta's acceptable-use policy prohibits tampering with the capture LED. It allows face or voice identification only for accessibility apps with Meta's prior approval [8]. Determine does no face identification. FACT

CHAPTER 11

HackTX 26 build plan

11.1 Event facts (official site)

- Hacking runs from **8 am Saturday 24 Oct to 5 pm Sunday 25 Oct 2026** on the UT Austin campus, fully in person. Teams have at most 4 people. The event is free and does not reimburse travel [32].
- Rule: **"All work must be done at the event. You can't demo something you didn't build."** Attendees follow the MLH Code of Conduct [32].
- Prizes exist for challenges, with details "revealed at the opening ceremony". No judging rubric, tracks or prize values are published [32].
- The listed sponsors include Meta (highlighted tile), Visa, Annapurna (linked to Annapurna Labs), NMC2, TFS, BHW, D. E. Shaw and Capital One [32]. HackTX's own announcement names Annapurna Labs its "first platinum partner" for HackTX 26 [68]; the HackTX 2025 Devpost listed no Annapurna sponsor or prize [69].
- Conflict on the site: the application banner says "Apply by Sept 15" while the FAQ says Sept 11. Walk-in admission is first come, first served, is not guaranteed, and the site discourages non-Austin attendees from travelling for it [32]. **Confirm acceptance in the portal before travelling.**

> **Implication:** all work, including this document, is done at the event: buy parts, read documentation, plan, write firmware, write the app and assemble the frame there. Email `hello@freetailhackers.com` to confirm whether cue sound files and phone compatibility checks are allowed. Get the answer in writing.

11.2 Team roles (4 people)

Role	Owns
Firmware	ESP32-CAM JPEG stream; XIAO ESP32-C3 single-zone ToF, v2 UDP protocol, haptic driver, haptic-only watchdog, heartbeat
Android	FrameSource and RangeSource ingest, ToF banding, HazardPolicy, latency logging
Audio / UX	Phone-side cue synthesis and headset routing (explicit failure), calibration flow, TalkBack-accessible UI, spearcons, judge diagnostics overlay
Hardware / demo	Frame mounting, wiring, power, the staged course, test protocol, Describe loop (Lambda API, Neuron model), pitch and Devpost

11.3 Timeline with abort branches

Window	Milestone	Abort or branch
Sat 8–12	Two-board bring-up: camera JPEG and scalar RANGE reach the phone; power and pin map checked. Resolve the motor shortage, regulator choice and integrated audio before a full wearable claim	Unsupported camera → JPEG at CIF or below [13], else phone-camera bench path. A borrowed compatible headset can exercise existing audio software only; it is not approval of the rejected headset build. No speaker fallback
Sat 12–18	End-to-end loop: ToF band → policy → phone audio + C3 haptic CUE. Timestamps logged	Camera not streaming → the hazard loop is unaffected; only Describe and frame diagnostics lose input

Window	Milestone	Abort or branch
Sat 18–24	Calibration flow on the headset, three modes, haptics, UNAVAILABLE handling	DRV2605L or mux trouble → re-seat the driver boards and re-check addresses; if haptics cannot be repaired, announce haptics unavailable (fault bits 3 or 4) and demo audio only, saying urgency then rides the audio alone. No substitute motor-drive parts are on the list
Sun 0–8	Gates G6–G8 on the staged course. Tune bands and rates. Freeze features	No new features after 8 am
Sun 8–12	Optional adapters: Describe loop end to end (only if accelerator quota was granted), Meta DAT through Mock Device Kit, Auracast demo broadcast (only on supported phones)	Skip if any core gate is failing
Sun 12–17	Pitch rehearsal, Devpost, demo resets, spare batteries	—

11.4 Three-minute judge demo

1. **Hook (20 s):** the parking-sensor line, moved from the bumper to the brow. Hold up the two-board frame and the headset.
2. **Live course (75 s):** a sighted spotter walks the wearer slowly toward a single large, light-coloured obstacle (a foam board or wall) head-on, then past a head-height foam bar. The judge diagnostics screen shows both streams' age, the single forward distance and its VALID / NO_TARGET / UNKNOWN status, the chosen band and measured latency. The demo never claims a side and never labels the ToF return with a camera object.
3. **One-ear mode (30 s):** switch the profile to SSD mode live. Headset audio goes mono; the temple haptics keep firing for every hazard. Show the phone-side ceiling stopping a forced level increase.
4. **Pull the plug (20 s):** unplug the camera board. The system-off cue plays within the measured bound; reconnect and wait for READY. No "clear" is ever heard.
5. **Ask (20 s):** after recovery, long-press "What's ahead?" If the Describe loop is ready, a redacted frame returns a two-sentence answer prefixed "may be wrong". If accelerator access failed, skip the answer and keep showing the offline warning loop.
6. **Close (15 s):** the logged p50/p95 latency, the miss and false-alert table, what it cannot see, and what comes next.

11.5 Sponsor alignment, without forcing it

Meta

Wearables DAT 1.0 (rollout from 30 Sep 2026) gives mobile apps camera, microphone, audio and IMU access on supported AI glasses. Its Mock Device Kit simulates the glasses camera without hardware [6]. Determine's `FrameSource` seam lets a DAT adapter feed the *same* hazard loop. The story: Determine runs on commodity dumb glasses *and* on Ray-Ban Meta. Claim it only if the adapter was built at the event.

Annapurna Labs (AWS)

Annapurna Labs is AWS's Austin-based chip group behind Nitro, Graviton, Inferentia and Trainium [33]. The honest connection is *system-first co-design*: Determine places computation in the frame, the phone or the cloud according to latency, power and failure budgets (Chapter 6), and measures them. The optional Describe loop targets that silicon: a Graviton2 Lambda API holds the prompt, and SmolVLM would run on Inf2 via Optimum Neuron, with Trainium2 the documented alternative (§7.3) [34][35][62]–[65]. They stay out of the hazard loop. Known collaborations and how to use them: §11.6.

11.6 Annapurna Labs collaborations (checked 24 Oct 2026)

Collaboration	What the source shows	Use for Determine
HackTX 26 platinum partner	Announced as the event's first platinum partner [68]. No Annapurna challenge, rubric or prize is published; challenge details come at the opening ceremony [32]. Annapurna did not appear on the HackTX 2025 sponsor or prize list [69]	Do not build for an assumed track. At opening, ask the Annapurna team what they will judge, then decide whether the Describe loop is worth Sunday-morning time
Cal Hacks 12.0 co-host (Oct 2025)	Annapurna co-hosted; the live prize page listed "Best Use of Annapurna" with the prize "TBD" [70], and the Devpost prize filters list no Annapurna prize [71]	Precedent only: an Annapurna prize may be defined late or judged outside Devpost INFERENCE
Hugging Face	Hugging Face says it has worked with the AWS chip teams since Inferentia1 and maintains Optimum Neuron [72]. Optimum Neuron lists SmolVLM and Gemma3 for image-text-to-text on Inf2, with vLLM support for listed LLMs [62]. Its Neuron DLAMI is free and pre-installs Optimum Neuron; Inferentia and Trainium quotas start at 0 [73]	The Describe deployment path: HF DLAMI on one inf2.xlarge, SmolVLM first, Gemma3 as the second listed candidate if SmolVLM fails to compile. Request quota early; approval can take hours
Build on Trainium	A $110M program: Amazon Research Awards calls give selected proposals Trainium credits and research-cluster access; named universities include MIT, CMU, UC Berkeley, UIUC, UCLA and UTS [74]. UT Austin is not on that list, but the UT Austin–Amazon Science Hub names Prof. Adam Klivans (UT CS) as its Build-on Trainium ambassador [76]	After HackTX only, through a faculty principal investigator. Determine's Describe loop is small-model inference, so a proposal would need a chip-level research question, not a product demo
AWS Trainium Frontier Competition	Model-and-kernel co-design on Trainium2; $25K / $10K / $5K prizes, co-publication with Annapurna researchers and a NeurIPS 2026 event [75]. Submissions closed 30 Sep 2026	Closed. Shows what Annapurna rewards: measured hardware-aware design, not logo use
UT Austin–Amazon Science Hub	Amazon-wide partnership since 2023 (information retrieval, ML, robotics) with PhD fellowships, research collaborations and events [76]	Faculty route for post-event research with blind and low-vision participants; not an Annapurna sponsorship

What to show Annapurna judges. The placement table (frame, phone, cloud) with each budget measured on the day; for Describe, compile time, p50/p95 request time on inf2.xlarge and cost per request at the listed hourly price, all labelled C0 only if logged. Never claim a partnership, endorsement or credits from Annapurna. DESIGN

Winning overall is an ambition, not something this document can engineer. What it can engineer is the set of things judges reward: a complete working loop, a clear human need backed by evidence, measured numbers, honest limits and a memorable demo.

CHAPTER 12

Micro Center-only sourcing and bill of materials

12.1 Where everything new is bought

One store. Every new purchase is a Micro Center item. Nothing ships from another retailer, no part comes from an electronics surplus shop or a big-box store, and no part waits on a courier. DESIGN

Store	Address	Hours / phone	Use
Micro Center Houston	5305 S Rice Ave, Houston, TX 77081	Mon–Sat 10–9, Sun 11–6 · (713) 940-8500 [38]	Stock snapshot below; shortages and audio block a complete shopping trip. Recheck before driving
Micro Center Austin	12707 N Mopac Expy, Austin, TX 78727	Opens 10 am Saturday [38]	Top-up during HackTX (hacking starts at 8 am)

Snapshot: 24 Oct 2026, ~1:40 PM CDT, real Zen browser, Houston selected. Counts and prices below are website observations, not reservations or receipts; they may change and exclude tax. Earlier automated HTTP 403 failures no longer describe this audit. No purchase was made. Nothing is already owned unless confirmed.

12.2 Complete purchase list

Not purchase-ready: two motors are required but only one is listed; the LM3671 is sold out; integrated bone-conduction audio has no approved component. Buy only confirmed required items not already owned. Candidate substitutions below are not adopted replacements.

Part and role	Buy	Source, status and constraint
Arduino Modulino Distance (VL53L4CD), one forward zone	1	992826 [40]: **2 in stock, $11.99**. Solderable pins/jumpers, no Qwiic harness. Sensor spec [39]
XIAO ESP32-C3 3-pack (one used, two spare)	1 pack	069005: **3 packs, $15.59 each**
ESP32-CAM pair (one used, one spare, not stereo)	1 pair	221739: **6 pairs, $17.99 each**. Check actual sensor, PSRAM and pinout at G0
DRV2605L haptic drivers	2	008607: **4, $7.99 each**
ERM vibration disc motors	2	850917: **1, $1.99 — short one**
TCA9548A mux	1	278499: **2, $10.99 each**
LM3671 peripheral regulator, if G0 shows it necessary	1 if needed	009191: **sold out** ($4.95 listed). TPS62827 candidate below, not an assumed substitution
FTDI programmer	1	489252: **2, $7.99 each**. Verify 3.3 V UART logic and USB connector before wiring
Power bank: Energizer 10,000 mAh, USB-C + dual USB-A	1	911677: **6, $19.99**. Lists simultaneous charging up to 3 devices; combined draw untested
C3 USB-C cable; FTDI Mini/Micro-B combination cable; cut 5 V power lead	1 each	481952: 1/$10.99; 684290: 1/$14.07; 422618: 5/$2.99. Match connectors; insulate cut lead
M/F and F/F jumpers; headers; 22 AWG wire	1 pack/spool each	787911: 2/$14.99; 847194: 4/$7.99; 999805: 6/$5.00; 798124: 5/$5.99. Include flash jumper

Part and role	Buy	Source, status and constraint
Perfboard; solder; heat-shrink	1 pack each	422790: 1/$4.99; 472068: 1/$5.99; 797464: 4/$4.99
Resistor assortment; red recording LEDs	1 pack each	078410: 8/$7.99; 797175: 3/$7.99. Confirm LED GPIO/current limiting
Possible safety-glasses frame; Velcro; mounting tape; cable ties	1 each	045609: 1/$3.49; 448159: 4/$14.99; 713578: 1/$9.99; 447631: 25+/$2.99. Dedicated mounts not verified; fabrication and fit checks required
Weller 60 W iron; Klein MM325 meter; Android phone, only if needed	1 each if not owned	705236: 1/$32.99; 777540: 2/$34.99. Optional Moto G06 014555: 2/$99.99; app/hotspot compatibility untested
Integrated bone-conduction output	Unresolved	**No approved part.** OpenRun [41] redirects to support and is not approved; Humbird says “No longer carried”. Searches did not identify a suitable stocked discrete transducer; this is not proof none exists

Candidates only: TPS62827 262121: 1/$7.99, 3.3 V/2 A, 3.4–5.5 V input; verify pinout/load. Modulino Vibro 992891: 4/$11.99; different module/interface, needs redesign verification. Neither is a drop-in promise. No complete-build total while motor/audio decisions remain open.

12.3 Tiers

Tier	Contents	Cost status	Use
Tier 0: phone-only bench	Owned Android phone and its camera. Existing headset audio can be exercised on compatible bench equipment, not approved integrated hardware	Unquoted	Software proof, not completed glasses
Tier 1: intended haptics + camera	C3, two drivers/motors, mux, camera, Modulino, power and frame	Blocked	Resolve second motor and any required regulator; no cue audio
Tier 2: integrated audio + haptics	Tier 1 plus an approved, verified integrated bone-conduction output. No part selected	Blocked	Not a purchase-ready build; no shipped fallback

Other listed products (24 Oct 2026): Ray-Ban Meta Gen 3 from $449 [42], Envision Ally Solos $699 [43], OrCam MyEye 3 Pro $4,250 [44], NOA $4,990 [46], .lumen €9,999 [47]. They do different jobs. Determine's total is unquoted, so no like-for-like cost-of-entry comparison is claimed.

12.4 Purchase and fit gates

1. **Recheck the snapshot:** only buy required §12.2 items not already owned. Resolve the second motor, regulator requirement and integrated audio before calling this a complete-build shopping list. No courier fallback; if the Modulino is unavailable, stop and re-plan.
2. **Audio design gate:** the separate OpenRun is not approved. Stocked headsets do not establish integrated transducer availability. Select and verify output, mounting, safe drive/ceiling and latency before an audio-wearable claim. Existing Bluetooth software remains a bench path, with explicit route failure and no speaker fallback.
3. **Before wiring:** mock up a single forward camera and ToF on a headband, with the C3 and haptic drivers on the rear strap and the battery in a pocket. If balance, fit or simultaneous power fails Gate G0, do not claim a

wearable build. The Modulino is a 1.2 m single-zone near-field demo sensor only; it is not a grid, a stereo pair or a lateral-range sensor.

CHAPTER 13

Verification gates and failure catalogue

Gate	Test	Pass criterion (PROPOSED unless stated)
G0	Bring-up on two boards: ESP32-CAM JPEG frames and the C3 reading the Modulino Distance at 15 Hz, with the pin map and both power runs fixed. Record the camera module's actual PSRAM and sensor	10 minutes, no resets, no corrupt JPEGs, no brown-outs when both radios transmit
G1	Both Wi-Fi producers stream to the team's phone over one hotspot simultaneously	≥ 10 fps QVGA/VGA JPEG sustained 10 min with the RANGE stream active. Per-board frame and range loss plus staleness logged
G2	USB UVC on the team's phone (optional path)	Enumerates via Camera2 or a user-space library. Otherwise mark it unsupported
G3	Cue path: phone synthesises a cue, the headset plays it, the C3 fires haptics from the same CUE	Headset route confirmed active; audio-leg and haptic-leg p95 measured separately; unplugging or pausing the headset yields an announced fault, never phone-speaker audio
G4	Output ceilings: audio ceiling on the phone; `haptic_max` on the board	Audio can never be driven past the ceiling by a slider or urgency step. A CUE requesting more haptic than `haptic_max` is lowered and ACKed APPLIED_CLAMPED. PROFILE `haptic_max` = 0 is rejected as BAD_CUE (1–255 only), so the watchdog LSS always has a non-zero level. Minimum 1 is a protocol rule, not a measured guarantee the wearer can feel it — perceptibility is checked in G9
G5	Calibration repeatability	Two runs within ±3 dB per side, same fit
G6	End-to-end latency on the course	p50/p95/max over ≥ 100 events, range-sample-to-haptic and range-sample-to-headset-audio reported separately. Target haptic p95 ≤ 250 ms (report the actual value either way)
G7	Fault injection: unplug or power down either board, kill app, Wi-Fi off, headset off or stolen, low battery	Unavailable announced ≤ 1 s on a surviving output (headset audio, temple haptics or phone vibration). Phone dead: the board's haptic-only LSS watchdog, which makes no sound. No stale cue after the fault
G8	Single-zone ranging on the staged course: head-on approach to a white board, a gray board and a dark cloth, from 1.5 m inward, 10 repeats each, indoors with bright-window and glass cases. Log VALID / NO_TARGET / UNKNOWN rates and the distance at first VALID	Table of first-VALID distance and miss/false-alert rate per target. FAR 900–1200, MID 500–899 and NEAR < 500 mm are tuned from it; the 1.2 m spec is not claimed as useful warning distance unless the table supports it. No lateral or object-class claim is tested because none is made
G9	Cue identification after 5 minutes of training; haptic perceptibility at `haptic_max` = 1	About 90% correct family and band, in ambient noise (the number is arbitrary and needs a pilot). Report whether the minimum haptic level is felt
G10	Masking check: localise real sounds with cues on versus off	Report the difference. Reduce cue duty if it is large [52]
G11	Thermal and battery over 30 minutes, headset and headband together	Touch temperature comfortable. Runtime measured

13.1 Failure catalogue

Failure	Detection	Response
Phone lacks external-camera HAL	G2	Wi-Fi path (default anyway)
ESP32-CAM module has no PSRAM or an unsupported sensor	G0 bring-up; module markings vs the esp32-camera lists [13]	Without PSRAM the driver supports only JPEG at CIF or lower [13]: drop resolution, else Tier 0 phone camera
PSRAM contention corrupts frames under Wi-Fi [13]	JPEG decode errors	Lower resolution or frame rate. Use JPEG only
Hotspot drops or the phone roams to another network	Heartbeat timeout on both boards	UNAVAILABLE, then auto-rejoin. Pin the SSID
One board drops off the hotspot while the other stays up	That board's BEAT times out	Fail loud: UNAVAILABLE, never a half-working "all clear". Auto-rejoin
Clock-offset estimate drifts; a stale frame or RANGE sample arrives	Freshness gate (offset-corrected age + receive age)	Discard the input, re-estimate from PING/PONG. Persistent staleness → UNAVAILABLE
No usable ToF return: empty/out-of-range view, sunlight, glass or dark surface	RANGE status 2 UNKNOWN	Announce DEGRADED, stop metric warning repeats, never call it far or clear. A fresh valid return re-enters through normal persistence
Unusual or unlabelled obstacle in the cone	G8 log	No effect on the hazard loop: it uses no object detector. The ToF band triggers "obstacle" if the object is in the cone
A2DP latency on the headset audio leg [16]	G3 measurement	Haptics carry urgency. Audio carries identity
Headset disconnects, or multipoint switches it to another device	Audio route change callback	Announce explicitly on haptics and phone vibration; never fall back to the phone speaker. Disable multipoint for the demo
Phone app killed, crashed or Wi-Fi off	Board: no PING for 800 ms (or none within 5 s of boot)	Phone-owned audio is gone with it, so the only announcement is the board's temple LSS haptic every 5 s at `haptic_max` (≥ 1), fault bit 7. This is a haptic-only last channel, not an audible alarm
Wearer cannot lateralise bone cues	Calibration verify step	SSD mode: mono audio. Hazard cues never carried a side, so haptics keep confirming every hazard
Brown-out when both boards transmit at once	Reset reason in heartbeat	Bulk capacitance, short separate power runs, power bank with adequate output

CHAPTER 14

Roadmap after HackTX

Track	Next step	Gate before claiming
User research	Co-design sessions with blind and low-vision travellers, O&M specialists, and people with unilateral hearing loss	Ethics approval. No blindfolded substitutes [2]
Hearing-aid output	Send cues directly to LE Audio hearing aids and earbuds (unicast, LC3), so wearers who already use hearing aids need no extra transducer [16][19]	Device support plus measured latency
Auracast	Encrypted broadcast so an O&M instructor or companion can monitor cues in training [17][18]	Supported phones only. Never the safety path
Meta DAT	Production DAT adapter on supported Ray-Ban and Oakley Meta models, with bone and haptic output kept on a separate accessory [6][7]	Measured stream latency (undocumented by Meta)
Sensing	A multi-zone ToF if one becomes available from the same store, enabling tested lateral claims; synchronised stereo or ESP32-P4 with high-speed USB. A downward-pitched sensor for drop-off *research*	Per-class recall data with O&M review
Hardware	Custom flex PCB in the temple arms, integrated battery with protection, IP rating	Electrical, battery and RF compliance testing
Single-board wearable	Re-combine camera and cues on one SoC when a camera-capable board (e.g. XIAO ESP32-S3 Sense [14]) is reliably obtainable; removes the second radio and the per-board clock alignment	Availability plus a G0-equivalent two-function bring-up
iOS	Wi-Fi FrameSource. Apple documents UVC external cameras for iPad [56], so treat iPhone USB cameras as unsupported until tested	Parity of gates G1–G7
Regulatory	Classification review under 21 CFR 886.5900 [31]	Legal counsel
Annapurna / AWS research	Ask the UT–Amazon Science Hub's Build-on Trainium ambassador whether a chip-level question fits an Amazon Research Awards proposal [74][76]	A faculty principal investigator and an accepted proposal

CHAPTER 15

Architecture decision records

ADR	Decision	Alternatives rejected	Revisit when
001	Phone is the only decision-maker	On-glasses inference (cost, heat, battery). Cloud inference (latency, offline failure)	A frame SoC can run the policy at similar cost and power
002	Mono RGB (frames for Describe and diagnostics only) + one forward ToF zone for v0, replacing the unobtainable 8×8 grid (24 Oct 2026)	8×8 grid (not purchasable under the Micro Center-only rule). Dual RGB stereo (sync and calibration; the second ESP32-CAM is a spare). Ultrasonic (wide beam, weak bearing)	A Micro Center-listed multi-zone ToF becomes available, or a lateral-range claim is needed and tested
003	Wi-Fi hotspot transport for frames	BLE video (bandwidth). USB UVC by default (support varies by phone maker)	Gate G2 passes on the target phone fleet
004	Cue audio synthesised on the phone and played on a finished bone-conduction headset; the glasses fire haptics only (24 Oct 2026, replaces on-board I²S audio)	On-board stereo amplifiers and raw transducers (shipped parts, skin-contact wiring). Phone speaker (public, and silent fallback is forbidden)	LE Audio unicast is widely available, or a Micro Center-listed raw transducer path is justified by a latency measurement
005	Hazard cues carry no direction in v0; urgency is redundant across headset audio and haptics; no panning-only cues	Inferring a side from one zone. HRTF spatial audio alone (4 of 13 struggled [2]; SSD cannot use it)	A lateral-capable sensor is built and gated
006	Calibration matches perceived level per session, with a hard ceiling enforced on the phone audio path and a separate `haptic_max` clamp (1–255) on the board	Fixed balance slider. Unlimited boost of the weak side. `haptic_max` = 0 (would silence the last local failure channel)	A transducer dose standard exists
007	No "clear" cue; silence means no report from a healthy system	Periodic "all clear" reassurance	Never
008	Cloud and generative description out of the hazard loop	LLM-driven navigation (22.2% wrong answers [30])	Never for hazard cues
009	Meta DAT is a FrameSource adapter, not a dependency	DAT-only build (cost, no haptics or bone output)	The team owns supported glasses
010	Android first	iOS first (no UVC, Classic audio only)	After gates G1–G7 pass on Android
011	Two Wi-Fi producers: XIAO ESP32-C3 cue board + separate ESP32-CAM camera board, no board-to-board bus (24 Oct 2026)	Single XIAO ESP32-S3 Sense (not obtainable for this build). Hardwired board-to-board bus (unverified wiring, no spare bus on the C3, new failure modes)	A camera-capable single board is reliably purchasable, or two-station hotspot throughput fails Gate G1

CHAPTER 16

Source register

Retrieved 24 Oct 2026, except [13], [36], [59] and the AWS entries re-read for the 24 Oct two-board revision, [62] plus the Annapurna collaboration entries [68]–[76] read 24 Oct 2026, and the Modulino/VL53L4CD/Shokz/Micro Center entries [20], [37], [39]–[41] read or searched 24 Oct 2026. Micro Center product pages returned HTTP 403 to automated reads, so [40] and [41] are catalog links, not verified prices or stock. Vendor pages state vendor claims. Product prices drift.

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